

Crash Claim vs. Data Reality: Ford & Fatal Crashes



The Claim in the Headlines

Recent headlines warned that "Ford vehicles are involved in the most fatal crashes." This implies Fords might cause more deadly accidents. Such a claim caught management's attention – is there really a Ford danger, or is it a data mirage? We turned to the data to find out.



Our Big Question

Key Question: Do Ford vehicles truly lead to more fatal crashes, or are other factors at play? We set out to investigate the claim by analyzing a large NYC crash dataset (~49,000 collision records) for patterns. The core question driving our project: "Are Fords inherently more dangerous, or is something else behind the higher fatal crash count?"



Hunches & Hypotheses

Our Expectation: We suspected correlation $\neq$ causation. If Fords show up often in fatal crashes, it might be because Fords are common on the road – not because the brand causes crashes. We hypothesized that driver behavior and conditions like who's driving and when would matter more than car make. For example, we expected risky driving factors – such as an unlicensed driver, male driver, or poor conditions (night or busy season) – to drive up fatal crash risk. In short, our theory: Ford's higher fatal crash count is likely explained by other risk factors, not by the Ford brand itself.

By the Numbers

Data at a Glance: Before modeling, we reviewed the raw data to understand the context:

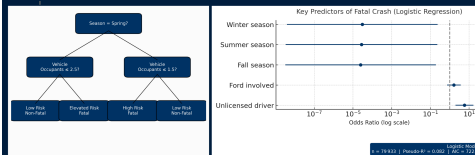
- **Tiny Fraction Fatal:** Out of ~49,000 NYC crashes, only 50 were fatal (~0.06%). Fatal accidents are extremely rare – a small data sample.
- **Ford's Presence:** Yes, Ford appeared in several of those 50 fatal crashes (the most of any make, as the media claimed). However, Ford is a popular make overall, so naturally it shows up often in crashes.
- **Seasonal Spike:** A pattern emerged with season – fatal crashes were slightly more frequent in summer months. Warmer seasons saw more deadly accidents than winter/fall.
- **Driver Factors:** We also noticed many fatal incidents involved male drivers and often multiple occupants in the vehicle. This suggested human factors might be behind the fatalities.



Deep Dive with Data Models

Our Approach: To test our hypotheses, we applied data mining models for clearer insights: a Decision Tree for interpretability and a Logistic Regression for validation.

- **Decision Tree Classifier:** We trained a decision-tree model to predict Fatal vs. Non-Fatal crashes. The tree would split on the most important variables first, effectively showing us which factors best "decide" fatal outcomes.
- **Feature Importance:** Our analysis showed that "Male Driver" (~25%), "1st of Occupants" (~24%), "Driver Licensed?" (~16%), "Summer Season" (~15%) were the top predictors, while "Ford Vehicle" ranked last at only ~5%.
- **Logistic Regression:** We also ran a logistic regression to double-check the findings. The results aligned with the tree: "Ford involvement" had a high p-value (insignificant, meaning that once we account for driver and situation variables, driving a Ford did not significantly raise the chance of a fatal crash).



Key Insights & Findings

Our models showed no special risk tied to the Ford brand. Instead, the top predictors of fatal crashes were:

- **Driver Factors:** Crashes were more likely to be fatal if the driver was male or driving without a valid license.
- **Multiple Occupants:** accidents with more occupants tended to result in fatalities more often.
- **Seasonal Context:** Timing played a role. Summer (and to a lesser degree spring) had a higher likelihood of fatal outcomes.
- **Vehicle Make:** Ford's ~5% importance in the model means it was almost irrelevant in predicting a deadly crash.



Bottom line: It's not about the car – it's about the driver and the context. The claim that "Ford causes more fatal crashes" is not supported when we look at the full data picture.

Caution: Know the Limits

Before we bang the gavel on our verdict, a few caveats:

- **Correlation vs. Causation:** This was an observational study, meaning we observed existing data without controlling variables. So, even when we see patterns, we cannot directly infer causation.
- **Data Limitations:** The dataset had only 50 fatal cases – a very small sample for modeling. With such rarity, predictions are difficult and come with high uncertainty.
- **Missing Factors:** We focused on key recorded variables. Other potential factors – like driver age, vehicle safety features, speed, road type, or how often each make is on the road – were not captured in our analysis.
- **Generality:** Our study used NYC crash data, which may have unique patterns. Results might differ in other regions or larger datasets.



The Verdict

Verdict: Ford is Not Guilty of causing more fatal crashes. The evidence points to a clear conclusion: vehicle make had little to no impact on crash fatality outcomes. The initial alarm about Ford was a classic case of mistaking correlation for causation.

What truly matters: Our findings suggest that to improve road safety, focus on the real risk factors – driver behavior and situational context. Male and unlicensed drivers, crowded vehicles, and seasonal spikes in crashes are the key issues to address, not the brand of car.

In conclusion, the data does not support the claim that "Ford causes more fatal accidents." It underscores a broader lesson: data-driven decisions require looking beyond surface statistics. By doing so here, we avoided a misdirected blame. Ford isn't the culprit – unsafe driving conditions are.

